

MANET - Routing Protocols

Module 41

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Video: <https://www.youtube.com/watch?v=9eNSUh66Zag>

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0:00 [Music]

0:20 Hello friends I am engineer s Kumar from school of electrical engineering and information technology Punjab Agriculture University Ludhiana I

0:29 welcome you all to the video lecture series of cyber and information security in this lecture we

0:36 will study about routing protocols used in mobile adog network so now let us

0:42 start before going into the details of routing protocol it is necessary to know about the routing protocol what it

0:51 performs actually and what is their importance in the field of communication generally routing protocol

0:59 represents the relationship or formula that is being used by routers to find the suitable way through which data can

1:07 be forwarded routing protocol is also helpful in exchange of information between routers routing protocol plays a

1:16 vital role to adjust the network according to variable conditions in other words the routing protocol is the

1:24 implementation of routing algorithm in the form of software or Hardware routing protocol uses different matrices to find

1:33 out the path which is used to transfer the packet across the network next types of routing

1:41 protocols there are three types of routing Protocols are used in mobile adhoc network reactive routing protocols

1:49 proactive routing protocols and hybrid routing protocols first reactive routing protocols reactive routing Protocols are

1:58 called On Demand routing protocols so these routing Protocols are called when they are needed and the routers are

2:07 built these routers can be acquired by sending root request through the network disadvantage of this algorithm is that

2:15 it offers High latency in searching a network next proactive routing protocols the routing information about all the

2:24 nodes is built and maintained by the proactive protocols the proactive routing Protocols are independent of

2:32 whether or not the route is needed control messages are transmitted with periodically intervals even if there is

2:40 no data flow still control messages are transmitted because these control messages proactive routing Protocols are

2:48 not bandwidth efficient there are many advantages and disadvantages of proactive routing protocols one of its

2:56 Advantage is that the nodes can easily get routing information and it easily starts a session the disadvantages are

3:06 too much data kept by the nodes for route maintenance when there is a particular

3:11 link Failure Its reform is too slow next hybrid routing protocol this approach is

3:19 a hybrid approach of combining both benefits of reactive and proactive routing protocols the routing initiates

3:27 using the proactive method of pre planned routes extra demand is then handled by reactive flooding this

3:35 methodology is only suitable for some situations where traffic demand and the number of nodes can be determined before

3:43 hand first protocol is aodv AD hoc on demand distance Vector this is reactive

3:50 protocol seek to set up routes on demand if a node wants to initiate communication with the node to which it

3:59 has no route the routing protocol will try to establish such a route aodv protocol is described in RFC

4:08 3561 the philosophy in aodv like all reactive protocols is that topology information is only transmitted by nodes

4:17 on demand when a node wishes to transmit traffic to a host to which it has no route it will generate a route request

4:26 which is rreq message that will be flooded in a limited way to other nodes this causes control traffic overhead to

4:35 be dynamic and it will result in an initial delay when it initiating such

4:41 communication a route is considered found when the RREQ reaches either the destination

4:49 itself or an intermediate node with a valid route entry for the destination for as long as route exists between two

4:58 end points if it remains passive when the route becomes invalid or lost aodv will

5:06 again issue a request aodv defines three types of control messages for route maintenance first

5:14 rrq a root request message is transmitted by a node requiring a route
5:20 to a node as an optimization aodv uses an expanding ring technique when
5:27 flooding these messages every rrq carries a time to live value that states
5:34 for how many hops this message should be forwarded this value is set to a
predefined value at the first
5:42 transmission and increased at retransmissions retransmissions occur if
5:48 no replies are received rrq frame format First Source address then Source
5:56 sequence then broadcast ID then destination address then destination
6:02 sequence Last Hope count next RRP a root reply message is
6:09 unicast back to the originator of rrq if the receiver is either the node
6:16 using the requested address or it has a valid route to the requested address the
reason one can unicast the message back
6:25 is that every route forwarding a rrq cache a route back to the originator the
6:32 format of RRP is Source address destination address destination sequence
6:39 hope count and lifetime where lifetime indicates how long a path can be used that
is r e r r
6:48 nodes monitor the link status of next stops in active routes when a link breakage in
an active route is detected
6:57 a root error message is used to notify other nodes of the loss of the link in order to
enable this report mechanism
7:06 each node keeps a precursor list which containing the IP addresses for each its
neighbors that are likely to use it as a
7:14 next hop towards each destination example the diagram to the left shows a
7:22 setup of four nodes on a wireless network the circles illustrate the range of
communication for each node
7:30 because of the limited range each node can only communicate with the nodes next
to it node one wishes to send a message
7:40 node three node one's neighbors are nodes 2 and four since node one cannot
7:47 directly communicate with node 3 node one sends a root request message the
7:54 root request is received by node four and node two when node one receives the root
request message they have two choices if
8:03 they know a route to the destination or if they are the destination they can send a
root reply message back to node
8:11 one otherwise they will rebroadcast the root request to their set of neighbors the
message keeps getting rebroadcast
8:20 until its lifespan is up if node one does not receive a reply in a set amount of time it
will rebroadcast plus the
8:29 request accepted this time the root request message will have a longer lifespan and a
new ID number all of the

8:37 nodes use a sequence number in the root request to ensure that they do not rebroadcast a root request in the

8:45 example node two has a root to node three and replies to the root request by sending out a root reply node four on

8:55 the other hand does not have a root to node three so it rebroadcast the root request in the example node one is the

9:04 forwarding a root reply to node four it notices that the root in the root reply has a better sequence number than the

9:12 root in its routing list node one then replace the root it currently has with

9:20 the root in the root reply next root error messages root error messages allow a to adjust root

9:29 Roots When nodes move around whenever a node receives root error it looks at the

9:35 routing table and removes all the roots that contain the bad nodes the diagram

9:43 to the left illustrates the three circumstances under which a node would broadcast a root error to its neighbors

9:51 in the first scenario the node receives a data packet that it is supposed to forward but it does not have a root to

9:59 the destination the real problem is not that the node does not have a root the problem is that some other nodes think

10:08 that the correct route to the destination is through that node in the second scenario the node

10:15 receives a root error that cause at least one of its roots to become invalidated if it happens the node would

10:24 then send out a root error with all the new nodes which are now unreachable in the third scenario the node detects that

10:32 it cannot communicate with one of its neighbors when this happens it looks at the root table for root that uses the

10:41 neighbors for a next hop and marks them as invalid then it sends out a root

10:48 error with the neighbors and the invalid Roots next routing protocol is

10:54 ospf ospf is a proactive routing protocol and is also called as table driven protocol because it permanently stores

11:03 and updates its routing table ospf keeps track of routing tables in order to

11:09 provide a route if needed ospf can be implemented in any ad hoc network due to

11:17 its nature ospf is called as proactive routing protocol this protocol works in collaboration with other nodes in Manet

11:25 through the exchange of topology information this exchange of information is done

11:31 periodically ospf uses multi-point links to avoid the broadcast of unnecessary packets

11:40 retransmissions in a network a node broadcast a message periodically to its

neighboring nodes this is done to

- 11:48 compute the multi-point relay set as well as the exchange of information about the neighborhoods from the information about
- 11:57 the neighborhood this node calculates the minimum set of one hop lay point that is needed to reach the two
- 12:05 hope neighbors and this set is called the multi-point relay set osr differs
- 12:12 from the link State protocols in two factors based on the dissemination of routing information first is by
- 12:20 construction that is only the multi-hop lay nodes of a node a need to forward updates about link state that are issued
- 12:28 by node a secondly the size of the link State update of node a is reduced because it only consist of those
- 12:37 neighbors that selected nodes a as their multi-point lay node thus we can conclude that olsr reduce the link State
- 12:46 protocol it is used in a network where nodes are densely deployed the olsr
- 12:53 calculates the shortest path in such networks to an arbitrary destination following are the functions of protocol
- 13:02 first neighbor sensing in wireless adog networks each node has to certify its nature of link with the neighbor nodes
- 13:11 because of radio transmission in osr specifications there are two types of Link symmetric and
- 13:20 asymmetric each node sends hello interval to its one of neighbors Hello message to achieve neighbor sensing
- 13:28 which have no not be forwarded when node send the message it contains node neighbor list and their link status
- 13:37 which allow them to reduce the hold to Hope neighbor and their status afterward MPR selection is made and the list is
- 13:47 added into hello messages in final step using this MPR list a MP selector list
- 13:54 is constructed which contains a neighbors list which have selected it as ampere now the messages received from
- 14:03 their ampere selectors will be forwarded by the nodes second ampere flooding the aim of multi-point relas is
- 14:12 possibly control traffic flooding in ampere flooding ampers are selected in such a way that when a flooding message
- 14:20 is transmitted by the Ampere set it must reaches all two hope neighbors Amper set
- 14:28 of a node n which is also represented as the smaller subset of symmetric one of
- 14:34 neighbors of n having symmetric links with two hope neighbors of Amper flooding mechanism makes way to the
- 14:42 elimination of transmission duplication as well as reception duplication is

minimized by its next topology diffusion

14:52 the objective of topology diffusion is to create routing tables using periodic topology control messages that is TC

14:59 messages TC messages are circulated by each node with the nonn empty amp selector set to all Network nodes

15:08 broadcasting at least links between itself and the nodes in its ampers Lector set to achieve topology diffusion

15:17 these TC messages contain sufficient information which enable nodes first to construct their topology table and then

15:25 to derive their routing table the roots are using short path algorithm such as digra shortest path algorithm after

15:34 their calculation and providing the best possible hopes number however there is always a need to recalculate the routing

15:42 tables to update the route information as the tables are based on information concerning links to neighbors and

15:51 topology which can be changed at any instant next protocol is Tora temporary order routing algorithm

15:59 Tora is a hybrid routing protocol it is effective in solving the existing limitations in Mobile AOG networks due

16:08 to the high mobility of nodes congestion is one of the major problems in manets traditional shortest path algorithm

16:16 adoptive shortest path algorithm and Link state routing can not work properly in Mobile Stations it is difficult to

16:24 update the routing tables of each Dynamic nodes in to each node broadcast a cury packet and

16:32 the recipients broadcast an update packet it Sports the loop free multiple route facilities using flat a non

16:42 hierarchical routing algorithm it also provides better scalability to discover a route it uses

16:48 a dag dag means directed a cyc graph and also uses a set of totally ordered

16:55 Heights at all times in this approach information may only in One Direction hence it is only unidirectional there is

17:04 no chance to fail in infinite low it performs three basic operations root Creations root maintainance and root

17:13 deletion to perform these three basic operations this protocol uses three control packets such as cury update and

17:23 clear first root creation two control packets such as cury and updates are used to create a new route within the

17:31 network a cury packet consist of a destination ID which is used for identifying the destination node and upd

17:40 packet holds destination ID and height of the node each node maintains a root
17:46 required flag which is initially unset and also maintains the time of broadcast for
the last update packet within a
17:56 network when a node has no directed link and an unset root required flag it
broadcast a cury packet to its neighbor
18:05 nodes and set its root required flag when any node receives a cury packet then it
checks the following condition
18:14 first if there is no down stream link and route required flag is unset it rebroadcast
the cury control packet and
18:23 sets the root required flag next if there is no Downstream link and root
18:30 required flag is set it discards the cury packet if the receiving node has at least one
Downstream link with null
18:38 height it sets its height and broadcast a upd packet if the receiving node has
18:46 at least one Downstream link with the nonnull height first compares the time of
Last Broadcast of update packet with
18:54 the time of Link over which cury packet was received became active next when the
link becomes active
19:02 in a node it broadcast a update packet and discards the query packet if the
19:09 root required flag is set of any node it broadcast a query packet when a node
receives a update package from its
19:17 neighbors it updates the entities of array height and proceeds through the
following steps if the root required
19:26 flag is set node sets its high height and update all the entities in its link
19:32 State array and unsets the root required flag then it broadcast a update packets
19:39 with New Height if the root required flag is not set it just update the entities of the
link State array and
19:49 applies the root maintenance techniques next a root maintenance in Mobile hog
Network Tora
19:57 maintains the root when any topological change has occurred it also ensur the
reestablishment of roots between nodes
20:06 within finite amount of time the root maintenance is needed for a node when its
height is non null if any neighbor
20:14 node has null value that neighbor node will not be considered for this operation
there are five different cases
20:22 available in Tora to root maintenance case one generate new reference level case
two propagate the highest neighbors
20:31 reference level case three reflect back a higher sub level case four partition
detection and case five is generate new
20:41 reference level in case one due to link failure the node lost its last downstep

20:48 link if the node has any upam neighbor node it updates its reference value
20:54 otherwise it sets its height to null if the node has no Downstream link it propagates the reference level to its
21:03 Upstream Neighbors which is show in case to due to link reversal if the
21:10 downstream links fail then the node reflects back the reference height with setting the reflection bit according to
21:17 the case three if the set reference value is equal to the reference height of neighbor nodes the nod has detected
21:25 the partition and has started to raise the root setting the height value to null which is shown in case four in case
21:34 five generation of new reference value occurs when the node has experienced a link failure between the time of
21:44 propagation of a reference level and reflects Su level next operation is root deletion in case four of root
21:52 maintenance node sets its height which is determined by the direction of edges to the destination and updates all the
22:02 entries of its link State array and broadcast control packet when a node receives a control packet it follows the
22:11 following procedures first if the reference level in the control packet is matched with the reference level of that receiving
22:20 node the node sets its height and sets the height for each neighbor to null it
22:27 also updat all the entries in the link State arrays and broadcaster control packet second if the reference level of
22:36 the control packet does not match with the reference level of the node it sets its height for each neighbor and it also
22:45 updates corresponding link State array entries at the end the height of each node in the portion of the network which
22:53 was partitioned is set to null and erase all the invalid r Roots so friends this
23:00 was all about roting protocols in Manet hope the concepts explained in this lecture were understandable and helpful
23:08 hope to see you in the next lecture till then goodbye thank you
23:14 [Music]